

Naïve Bayes and Logistic Regression

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A photograph of a wooden floor with a mattress placed on it. The mattress is shaped into a complex, winding, and convoluted path, resembling a maze or a highly irregular line. This visual metaphor is used to represent overfitting in machine learning, where a model is too closely tailored to a specific dataset and thus fails to generalize to new data.

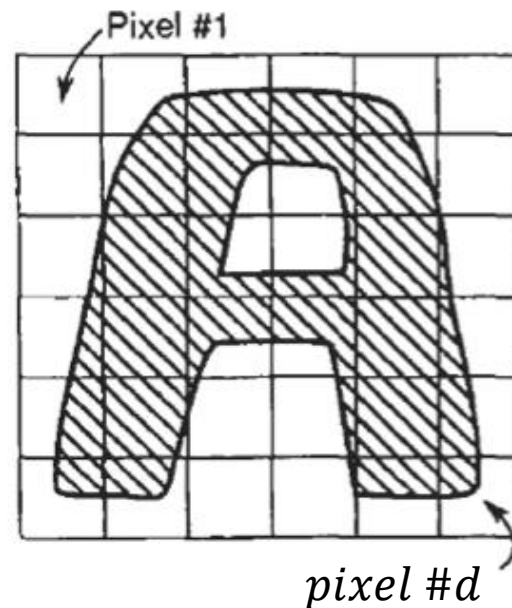
**THE BEST WAY TO
EXPLAIN OVERFITTING**

Outline

- Generative and Discriminative Classification 
- The Logistic Regression Model
- Understanding the Objective Function
- Gradient Descent for Parameter Learning
- Multiclass Logistic Regression

Classification

- Represent the data

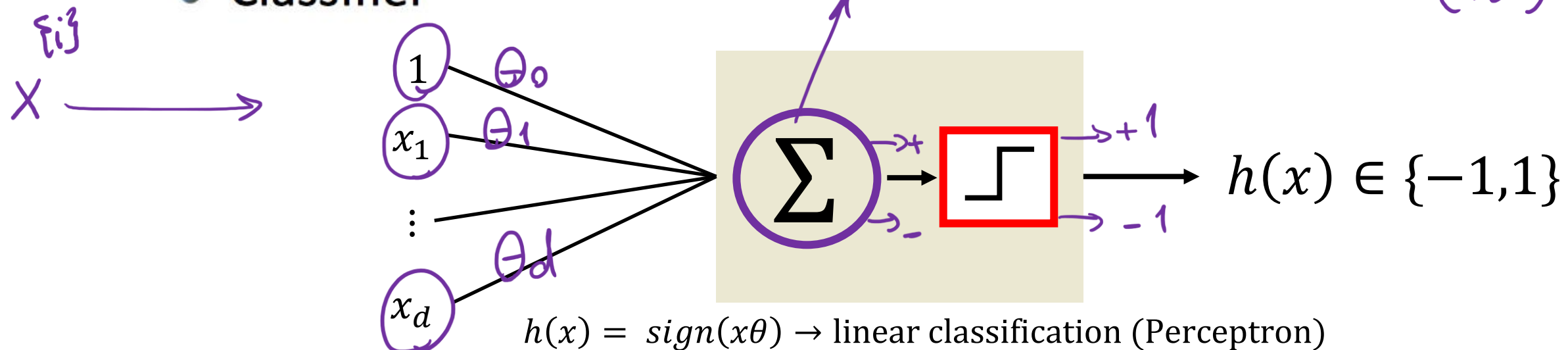


$$\Rightarrow X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

- A label is provided for each data point, eg., $y \in \{-1, +1\}$

$$\Sigma = LCF = \theta_0 + \theta_1 x_1 + \dots + \theta_d x_d = X\theta \in \mathbb{R} \quad (-\infty, +\infty)$$

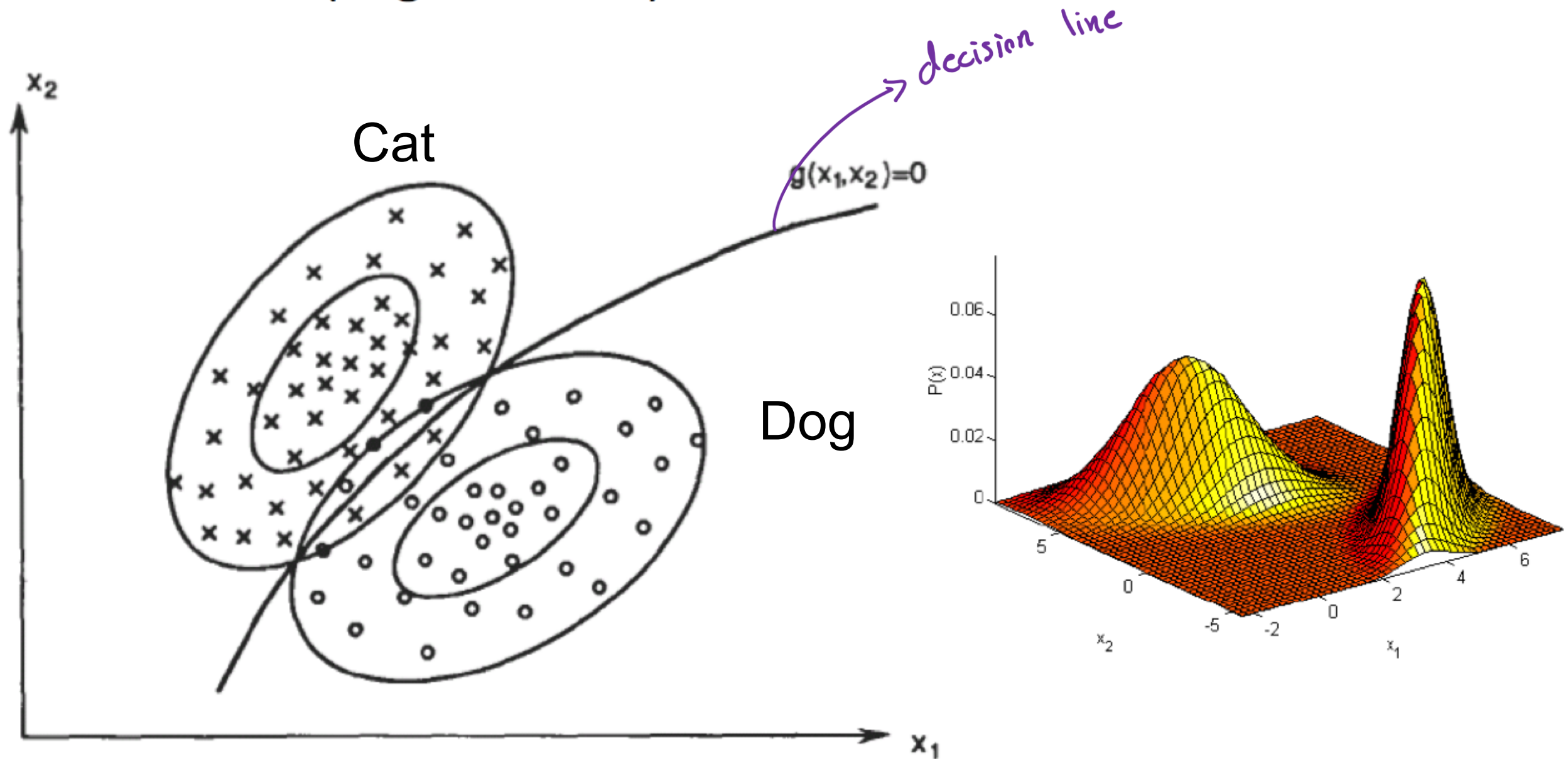
- Classifier



$$h(x) = \text{sign}(x\theta) \rightarrow \text{linear classification (Perceptron)}$$

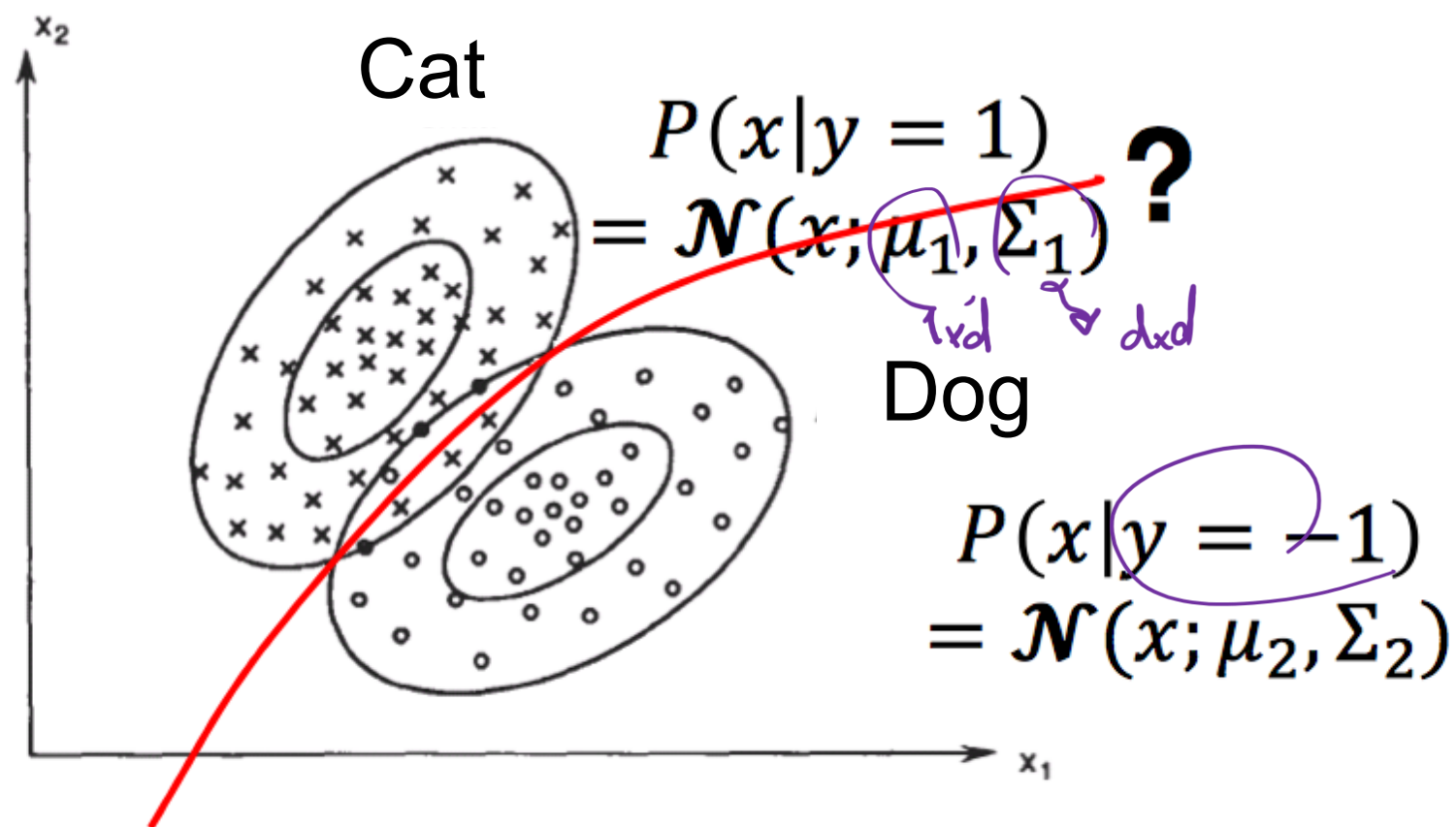
Decision Making: Dividing the Feature Space

- Distributions of sample from normal (positive class) and abnormal (negative class) tissues

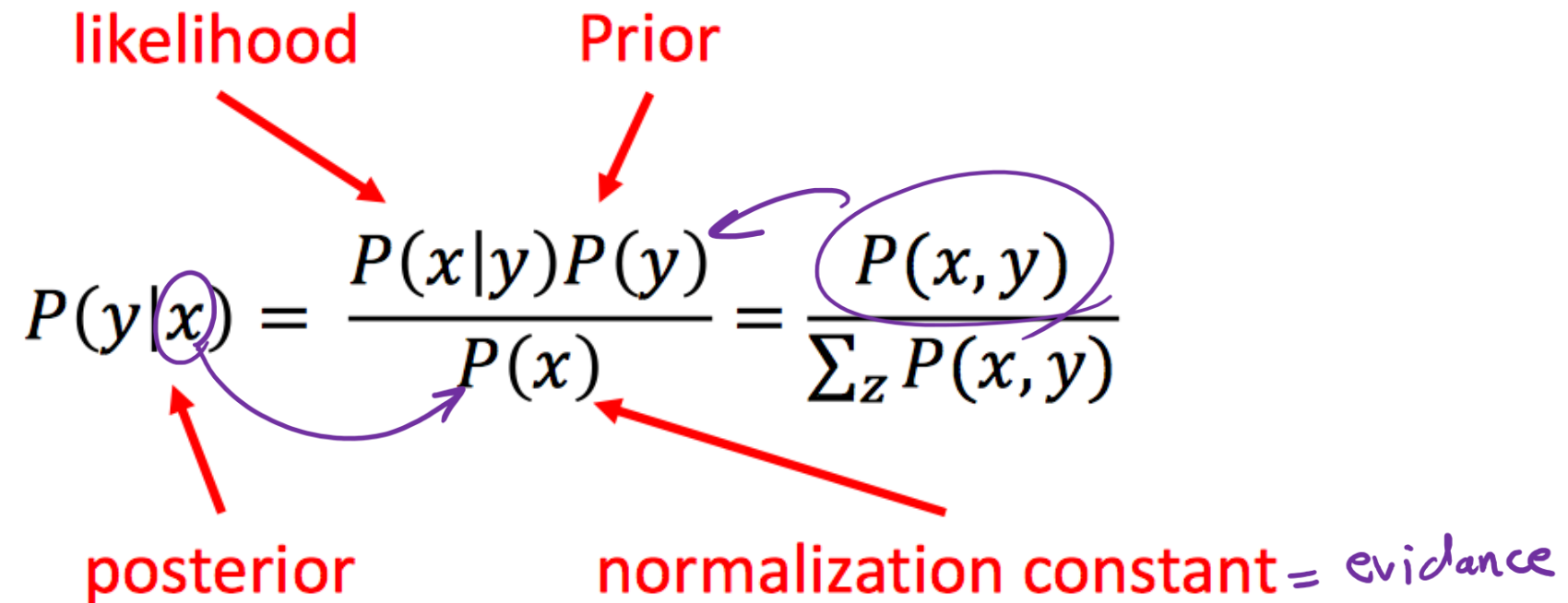


How to Determine the Decision Boundary?

- Given class conditional distribution: $P(x|y = 1), P(x|y = -1)$, and class prior: $P(y = 1), P(y = -1)$



Bayes Decision Rule



The diagram shows the equation for Bayes' theorem with several annotations. Red arrows point from labels to terms in the equation: 'likelihood' points to $P(x|y)$, 'Prior' points to $P(y)$, 'posterior' points to $P(y|x)$, and 'normalization constant = evidence' points to $P(x)$. Purple circles highlight $P(y|x)$, $P(x,y)$, and $P(x)$. A purple arrow points from the circled $P(x,y)$ to the circled $P(x)$.

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)} = \frac{P(x,y)}{\sum_z P(x,y)}$$

likelihood Prior

posterior normalization constant = evidence

Prior: $P(y)$

Likelihood (class conditional distribution : $p(x|y) = \mathcal{N}(x|\mu_y, \Sigma_y)$

$$\text{Posterior: } P(y|x) = \frac{P(y)\mathcal{N}(x|\mu_y, \Sigma_y)}{\sum_y P(y)\mathcal{N}(x|\mu_y, \Sigma_y)}$$

Bayes Decision Rule

- Learning: prior: $p(y)$, class conditional distribution : $p(x|y)$

- The poster probability of a test point

$$q_i(x) := P(y = i|x) = \frac{P(x|y)P(y)}{P(x)}$$

- Bayes decision rule:

- If $q_i(x) > q_j(x)$, then $y = i$, otherwise $y = j$

- Alternatively:

- If ratio $l(x) = \frac{P(x|y=i)}{P(x|y=j)} > \frac{P(y=j)}{P(y=i)}$, then $y = i$, otherwise $y = j$

- Or look at the log-likelihood ratio $h(x) = -\ln \frac{q_i(x)}{q_j(x)}$

$$P(Y=+1 | x^{i3}) = 1 - P(Y=-1 | x^{i3})$$

$$\underbrace{P(Y=+1 | x^{i3})}_{\text{Posterior}} = \frac{\underbrace{P(x^{i3} | Y=+1)}_{\text{Likelihood}} \underbrace{P(Y=+1)}_{\text{prior}}}{P(x^{i3})} = \frac{N(x^{i3} | \mu, \Sigma, Y=+1) P(Y=+1)}{P(x^{i3})}$$

$$P(x^{i3}) = \sum_{y=\pm 1} P(x^{i3}, Y=y) = P(x^{i3}, Y=+1) + P(x^{i3}, Y=-1)$$

$$= \frac{P(x^{i3} | Y=+1) P(Y=+1)}{a} + \frac{P(x^{i3} | Y=-1) P(Y=-1)}{b}$$

$$X = \begin{bmatrix} x^{i3} \\ \vdots \end{bmatrix}$$

$$N(x^{i3} | \mu, \Sigma, Y=+1) = \frac{1}{\sqrt{2\pi} |\Sigma|^{d/2}} \exp\left(-\frac{1}{2} (x^{i3} - \mu) \Sigma^{-1} (x^{i3} - \mu)^T\right)$$

$$X_1 \perp X_2 \perp X_3 \dots \perp X_d$$

$$P(x_1^{i_1}, x_2^{i_2}, x_3^{i_3}) = P(x_1 | x_2, x_3) P(x_2, x_3) = P(x_1 | x_2, x_3) P(x_2 | x_3) P(x_3)$$

$$P(x_1^{i_1}, x_2^{i_2}, x_3^{i_3} | y=+1) = P(x_1 | x_2, x_3, y=+1) P(x_2 | x_3, y=+1) P(x_3 | y=+1)$$

$$= (P(x_1^{i_1} | y=+1) P(x_2^{i_2} | y=+1) P(x_3^{i_3} | y=+1))$$

$$= N(x_1^{i_1} | \mu, \Sigma, y=+1) \dots N(x_d^{i_d} | \mu, \Sigma, y=+1)$$

$$P(y=+1 | x^{(i)}) = \frac{P(x_1^{(i)}, x_2^{(i)}, \dots, x_d^{(i)} | y=+1) P(y=+1)}{P(x^{(i)})}$$

$$= \frac{P(x_1^{(i)} | y=+1) P(x_2^{(i)} | y=+1) \dots P(x_d^{(i)} | y=+1) P(y=+1)}{P(x^{(i)})}$$

We assume

feature are ^{conditional} independent of each other

What do People do in Practice?

- Generative models
 - Model prior and likelihood explicitly
 - “Generative” means able to generate synthetic data points
 - Examples: Naive Bayes, Hidden Markov Models
- Discriminative models
 - Directly estimate the posterior probabilities
 - No need to model underlying prior and likelihood distributions
 - Examples: Logistic Regression, SVM, Neural Networks

$$P(y=+1 | x^{(i)}) = ?$$

Generative Model: Naive Bayes

- Use Bayes decision rule for classification

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)}$$

- But assume $p(x|y = 1)$ is fully factorized: ^{Conditional} Dimensions are independent.

$$p(x|y = 1) = \prod_{i=1}^d p(x_i|y = 1) = p(x_1|y=1) \dots p(x_d|y=1)$$

- Or the variables corresponding to each dimension of the data are independent given the label

“Naïve” conditional independence assumption

$$P(y|x) = \frac{P(x|y)P(y)}{P(x)} = \frac{P(x, y)}{P(x)}$$

Joint probability model:

$$P(x, y_{label=1}) = P(x_1, \dots, x_d, y_{label=1}) = P(x_1 | x_2, \dots, x_d, y_{label=1}) P(x_2, \dots, x_d, y_{label=1})$$


$$= P(x_1 | x_2, \dots, x_d, y_{label=1}) P(x_2 | x_3, \dots, x_d, y_{label=1}) P(x_3, \dots, x_d, y_{label=1})$$
$$= \dots$$

$$= P(x_1 | x_2, \dots, x_d, y_{label=1}) P(x_2 | x_3, \dots, x_d, y_{label=1}) \dots P(x_{d-1} | x_d, y_{label=1}) P(x_d | y_{label=1}) P(y_{label=1})$$

Naïve Bayes assumption: let's rewrite it as:

$$P(x, y_{label=1}) = P(x_1 | y_{label=1}) P(x_2 | y_{label=1}) \dots P(x_d | y_{label=1}) P(y_{label=1}) =$$

$$P(y_{label=1}) \prod_{i=1}^d P(x_i | y_{label=1})$$

Gaussian naïve Bayes
A typical assumption

Example

Discriminative Models

- Directly estimate decision boundary $h(x) = -\ln \frac{q_i(x)}{q_j(x)}$ or posterior distribution $p(y|x)$
 - Logistic regression, Neural networks
 - Do not estimate $p(x|y)$ and $p(y)$
 - Why discriminative classifier?
 - Avoid difficult density estimation problem 
 - Empirically achieve better classification results
- Generative model

Outline

- Generative and Discriminative Classification
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Gaussian Naïve Bayes

$$P(y = 1|x) = \frac{P(x|y = 1)P(y = 1)}{P(x)} = \frac{P(y = 1) \prod_{i=1}^d P(x_i|y = 1)}{P(x)}$$

$$\begin{aligned} \prod_{i=1}^d p(x_i|y = 1, \mu_{1i}, \sigma_{1i}) \\ = \prod_{i=1}^d \frac{1}{\sqrt{2\pi}\sigma_{1i}} \exp\left(-\frac{1}{2\sigma_{1i}^2} (x_{1i} - \mu_{1i})^2\right) \end{aligned}$$

Prior: $p(y = 1) = \pi_1$

Posterior: $p(y = 1 | x, \mu, \sigma, \pi)$

$$\frac{a}{a+b} = \frac{\pi_1 \prod_{i=1}^d \frac{1}{\sqrt{2\pi}\sigma_{1i}} \exp\left(-\frac{1}{2\sigma_{1i}^2} (x_i - \mu_{1i})^2\right)}{\sum_{\substack{k=1 \\ \text{labels}}}^2 \pi_k \prod_{i=1}^d \frac{1}{\sqrt{2\pi}\sigma_{ki}} \exp\left(-\frac{1}{2\sigma_{ki}^2} (x_i - \mu_{ki})^2\right)}$$

get $\exp(\ln(u))$ of numerator and denominator

$$= \frac{\exp\left(-\sum_{i=1}^d \left(\frac{1}{2\sigma_{1i}^2} (x_i - \mu_{1i})^2 + \log \sigma_{1i} + C\right) + \log \pi_1\right)}{\sum_{k=1}^2 \exp\left(-\sum_{i=1}^d \left(\frac{1}{2\sigma_{ki}^2} (x_i - \mu_{ki})^2 + \log \sigma_{ki} + C\right) + \log \pi_k\right)}$$

$$\frac{\frac{a}{a+b}}{\frac{a}{a+b}} = \frac{\frac{1}{1+\frac{b}{a}} \exp \left(-\sum_{i=1}^d \left(\frac{1}{2\sigma_i^2} (x_i - \mu_{1i})^2 + \log \sigma_i + C \right) + \log \pi_1 \right)}{\sum_{k=1}^2 \exp \left(-\sum_{i=1}^d \left(\frac{1}{2\sigma_i^2} (x_i - \mu_{ki})^2 + \log \sigma_i + C \right) + \log \pi_k \right)}$$

$$= \frac{1}{1 + \exp \left(-\underbrace{\sum_{i=1}^d \left(\underbrace{x_i}_{x_i} \underbrace{\frac{1}{\sigma_i} (\mu_{1i} - \mu_{2i})}_{\theta_i} \right)}_{\theta_0} + \underbrace{\frac{1}{\sigma_i^2} (\mu_{1i}^2 - \mu_{2i}^2)}_{\theta_0} \right) + \log \frac{\pi_2}{\pi_1}}$$

$$\sum_i \theta_i x_i = x_1 \theta_1 + x_2 \theta_2 + \dots + x_d \theta_d + \theta_0 = X \theta \rightarrow \text{LCF}$$

$$1 + \exp(-X \theta)$$

$$P(y = 1|x) = \frac{1}{1 + \exp \left(- \sum_{i=1}^d \left(x_i \frac{1}{\sigma_i} (\mu_{1i} - \mu_{2i}) + \frac{1}{\sigma_i^2} (\mu_{1i}^2 - \mu_{2i}^2) \right) + \log \frac{\pi_2}{\pi_1} \right)}$$

Number of parameters:

$2d + 1 \rightarrow d$ mean, d variance, and 1 for prior

$$P(y = 1|x) = \frac{1}{1 + \exp[-(\sum_{i=1}^d (\theta_i x_i) + \theta_0)]} = \frac{1}{1 + \exp(-s)}$$

(Handwritten notes: A purple circle around $P(y = 1|x)$ with an arrow pointing to the simplified equation. A purple '0' is written above the simplified equation, and a purple '1' is written to the right of the simplified equation. A purple arrow points from the '0' to the '1'.)

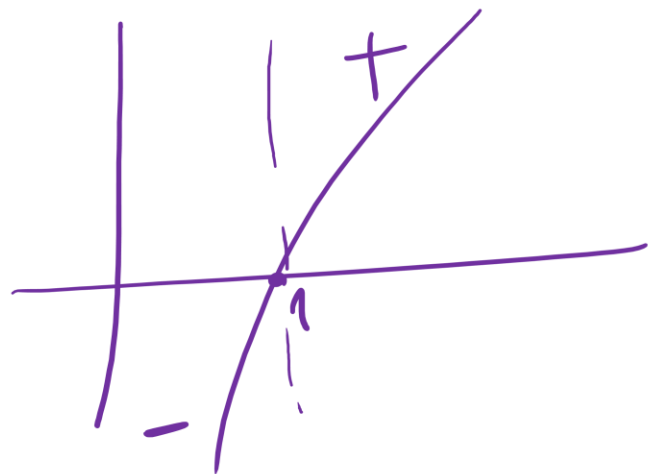
Number of parameters = $d + 1 \rightarrow \theta_0, \theta_1, \theta_2, \dots, \theta_d$ *(Handwritten note: $s = LCF = x\theta$)*

Why not directly learning $P(y = 1|x)$ or θ parameters?

Gaussian Naïve Bayes is a subset of logistic regression

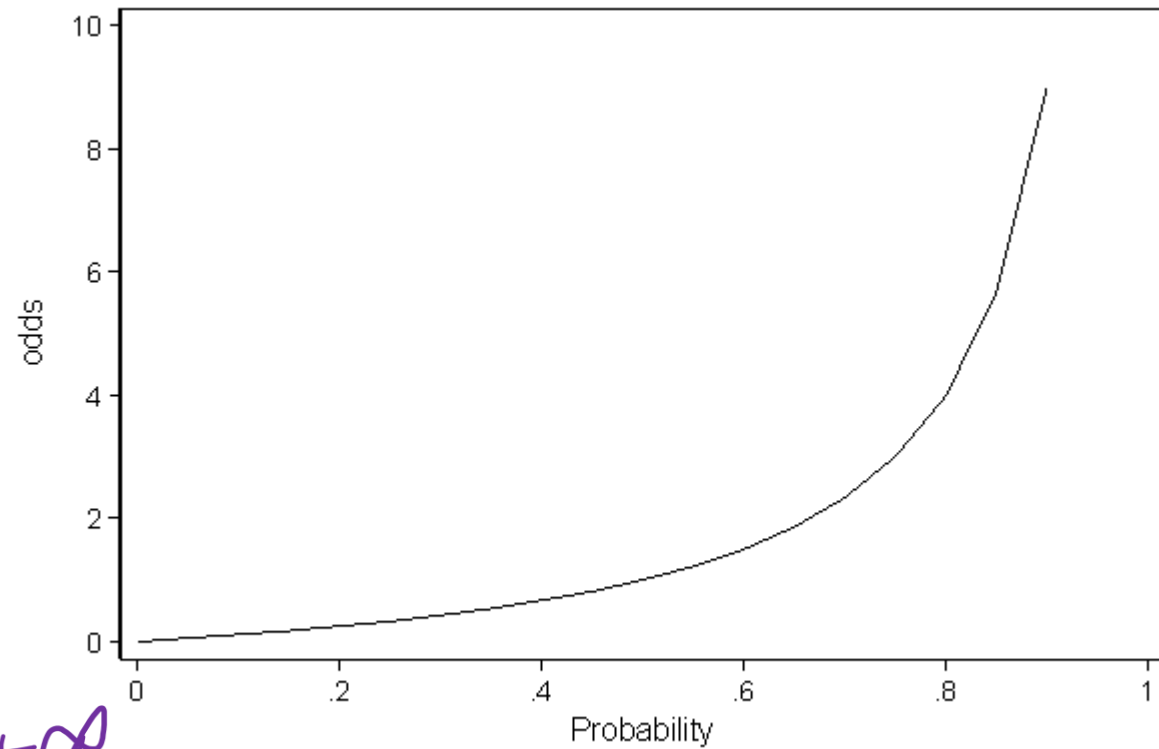
Why $\frac{1}{1+\exp(-x\theta)}$ is a probability?

$\frac{P(y = 1|x)}{1-P(y = 1|x)}$ is called Odds
 0.0001
 0.99

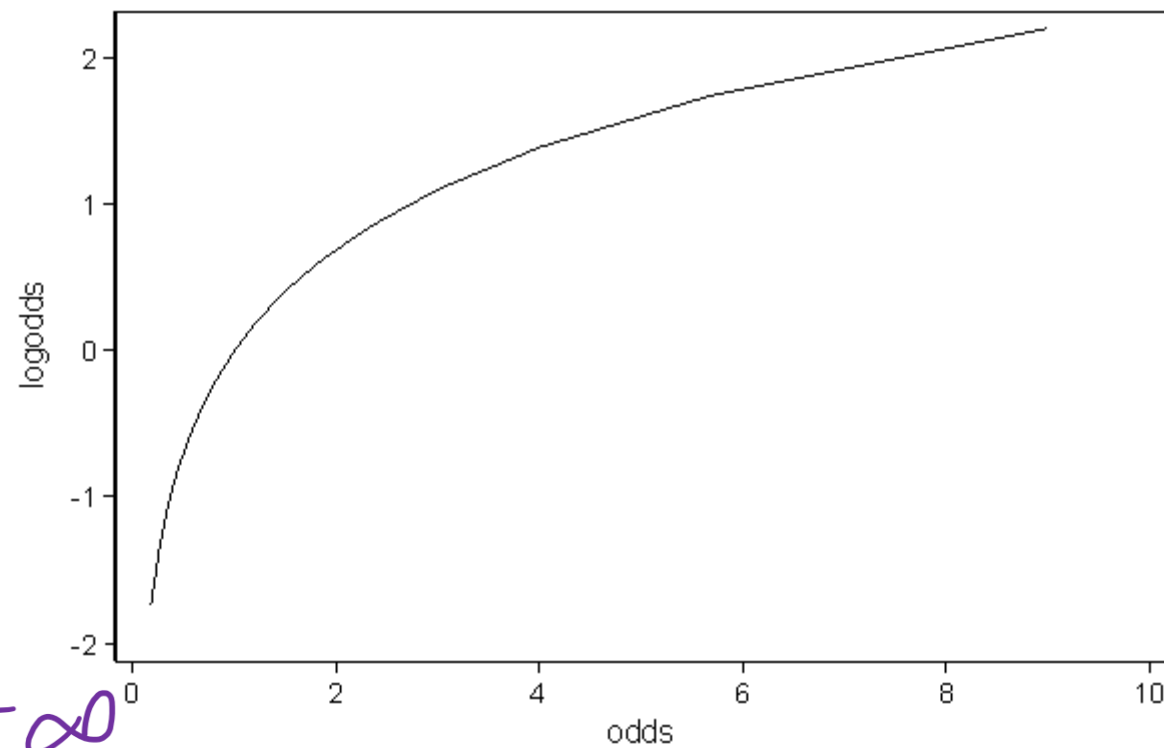


log(odds) vs odds

What could be $x\theta$ domain?



$+\infty$



$-\infty$

What is logit function?

$$P(Y=1 | x^{(i)}) = \frac{1}{1 + \exp(-(x\theta))}$$

$$\text{logit}(p) = \log(\text{odds}) = \log\left(\frac{p}{1-p}\right)$$

$$\log\left(\frac{p}{1-p}\right) = x\theta \Rightarrow \frac{p}{1-p} = \exp(x\theta)$$
$$\exp\left(\log\left(\frac{p}{1-p}\right)\right) = \exp(x\theta)$$

$$\log\left(\frac{p}{1-p}\right) = \theta_0 + \theta_1 x_1 + \dots + \theta_d x_d = \sum_{i=0}^d x_i \theta_i = x\theta$$

$$\exp\left(\log\left(\frac{p}{1-p}\right)\right) = \exp(x\theta)$$

Sigmoid

$$p = \frac{e^{x\theta}}{1 + e^{x\theta}} = \frac{1}{1 + e^{-x\theta}}$$

$$0 < \frac{1}{1 + \exp(-x\theta)} < 1$$

Logistic function for posterior probability

$$P(Y=1|x^{(i)}) \quad P(Y=0|x^{(i)}) = 1 - P(Y=1|x^{(i)})$$

$$P(Y=1|x^{(i)}) = \frac{1}{1 + \exp(-x^{(i)\top}\theta)}$$

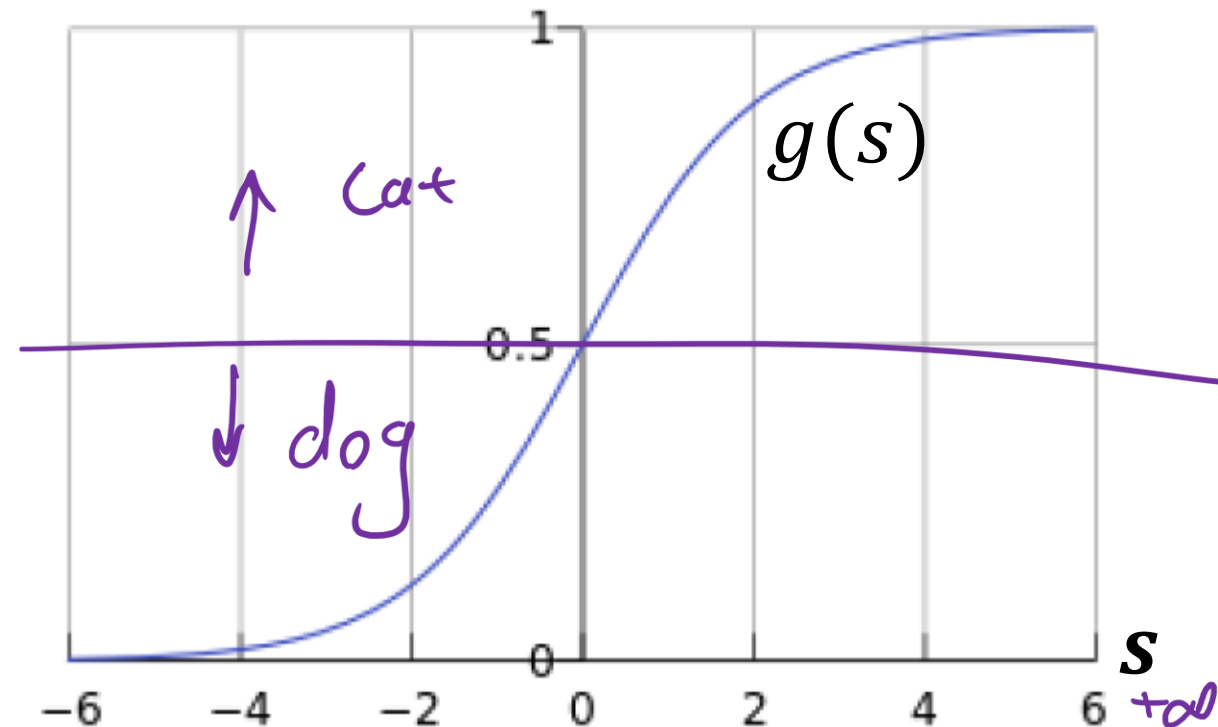
Let's use the following function:

$$(s) = x\theta$$

$$g(s) = P(y = 1|x) = \frac{e^s}{1 + e^s} = \frac{1}{1 + e^{-s}}$$

This formula is called sigmoid function $-\infty$

Many equations can give us this shape



It is easier to use this function for optimization

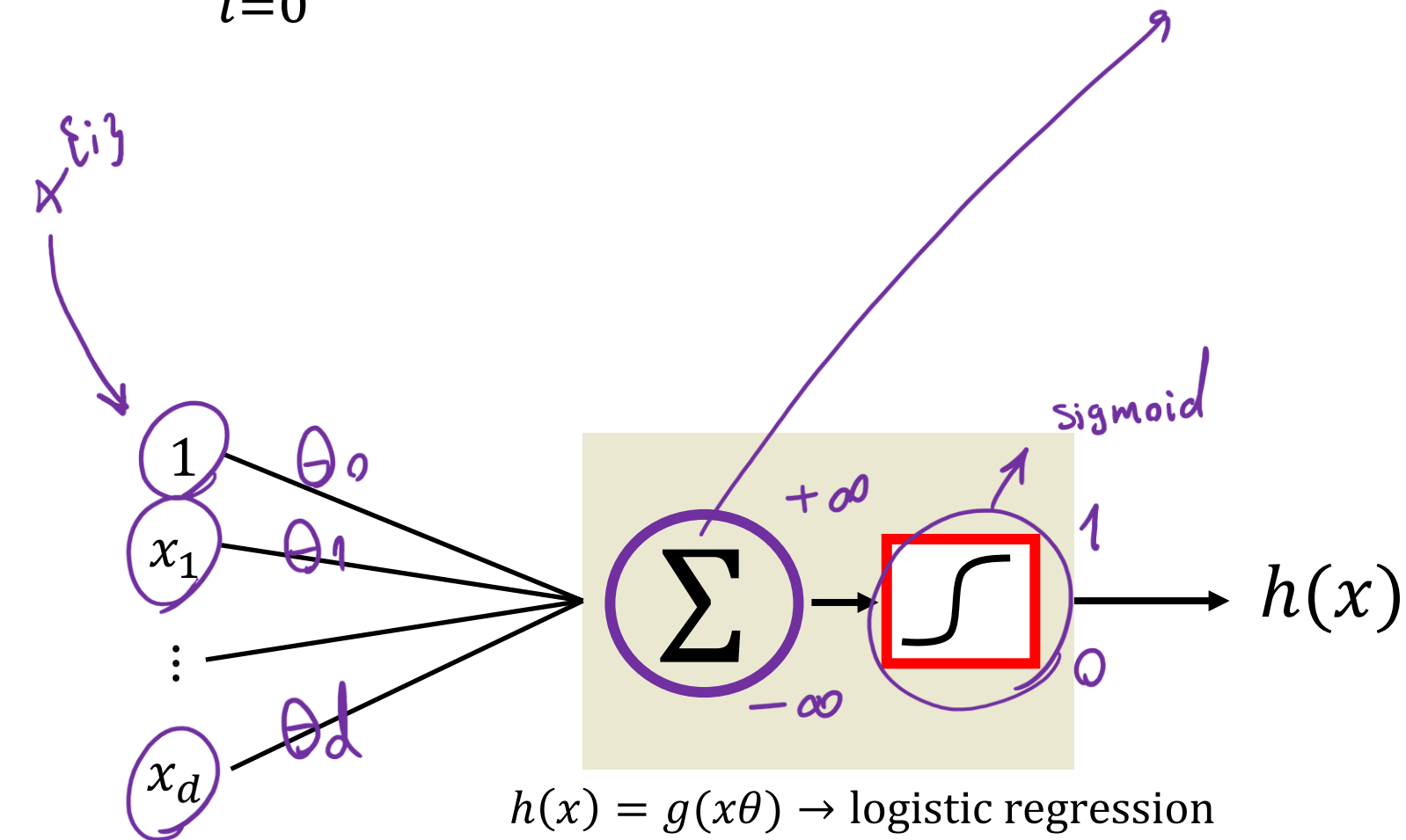
Is 0.5 threshold cut-off a good choice?

Learn about ROC and AUC (False positive rate and True positive rate) (Interactive)

$$g(s) = \frac{e^s}{1 + e^s} = \frac{1}{1 + e^{-s}}$$

Sigmoid Function

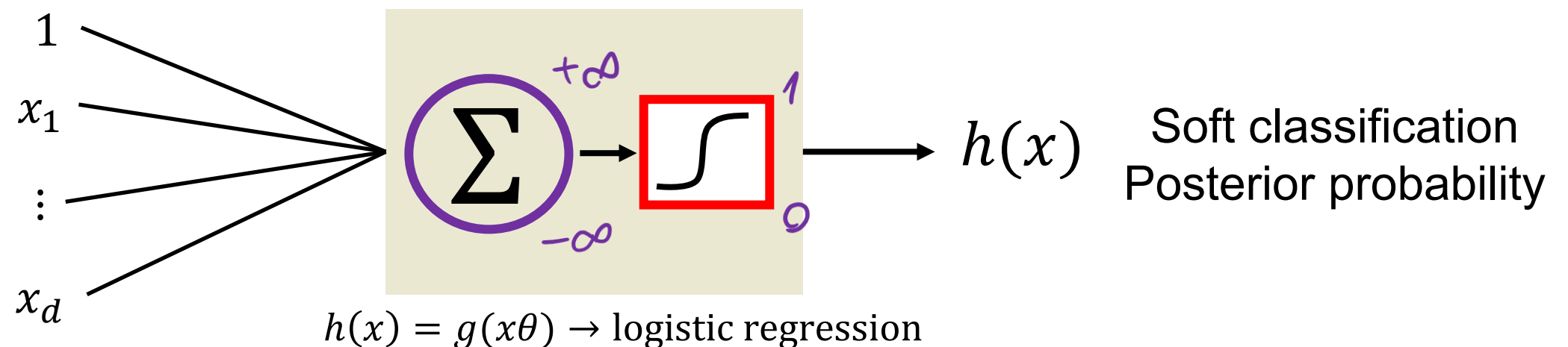
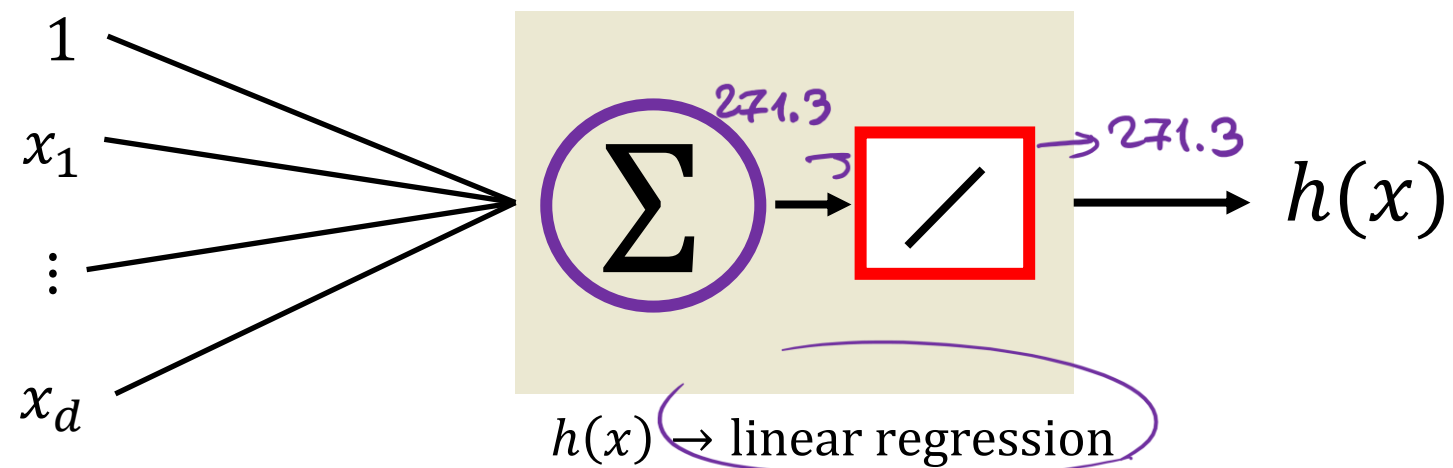
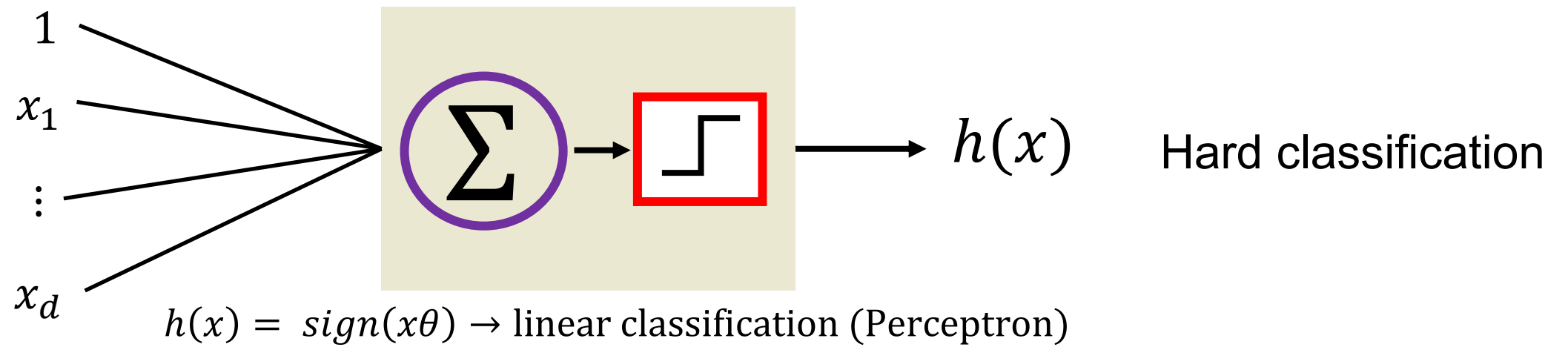
$$s = \sum_{i=0}^d x_i \theta_i = \theta_0 + \theta_1 x_1 + \dots + \theta_d x_d = LCF = x\theta$$



Soft classification
Posterior probability

$$s = \sum_{i=0}^d x_i \theta_i = \theta_0 + \theta_1 x_1 + \dots + \theta_d x_d$$

Three linear models



$g(s)$ is interpreted as probability

Example: Prediction of heart attacks

Input x : cholesterol level, age, weight, finger size, etc.

$g(s)$: probability of heart attack within a certain time

We can't have a hard prediction here

$s = x\theta$ Let's call this risk score

$$L(\theta) = \prod_{i=1}^N p(y^{(i)} | x^{(i)}, \theta) \Rightarrow \ell(\theta) = \log \prod_{i=1}^N p(y^{(i)} | x^{(i)}, \theta) \Rightarrow \frac{\partial \ell(\theta)}{\partial \theta} = 0$$

$$h_{\theta}(x) = \underline{\underline{p(y|x)}} = \begin{cases} g(s), \\ 1 - g(s), \end{cases} \quad \begin{matrix} \nearrow \text{cat} \\ y = 1 \\ y = 0 \end{matrix}$$

Using posterior probability directly

Logistic regression model

$$p(y|x) = \begin{cases} \frac{1}{1 + \exp(-x\theta)} & y = 1 \\ 1 - \frac{1}{1 + \exp(-x\theta)} = \frac{\exp(-x\theta)}{1 + \exp(-x\theta)} & y = 0 \end{cases}$$

We need to find θ parameters, let's set up log-likelihood for n datapoints

$$l(\theta) := \log \prod_{i=1}^n p(y^{(i)} | x^{(i)}, \theta) = \log \prod_{i=1}^n p(y^{(i)} | x^{(i)}, \theta)^{y^{(i)}} (1 - p(y^{(i)} | x^{(i)}, \theta))^{1-y^{(i)}}$$

$$= \sum_i \theta^T (x^{(i)})^T (y^{(i)} - 1) - \log(1 + \exp(-x^{(i)} \theta))$$

This form is concave, negative of this form is convex

The gradient of $l(\theta)$

$$l(\theta) = \log \prod_{i=1}^n p(y^{\{i\}} | x^{\{i\}}, \theta)$$
$$= \sum_i \theta^T (x^{\{i\}})^T (y^{\{i\}} - 1) - \log(1 + \exp(-x^{\{i\}} \theta))$$

- Gradient

$$\frac{\partial l(\theta)}{\partial \theta} = \sum_i (x^{\{i\}})^T (y^{\{i\}} - 1) + (x^{\{i\}})^T \frac{\exp(-x^{\{i\}} \theta)}{1 + \exp(-x^{\{i\}} \theta)} = 0$$

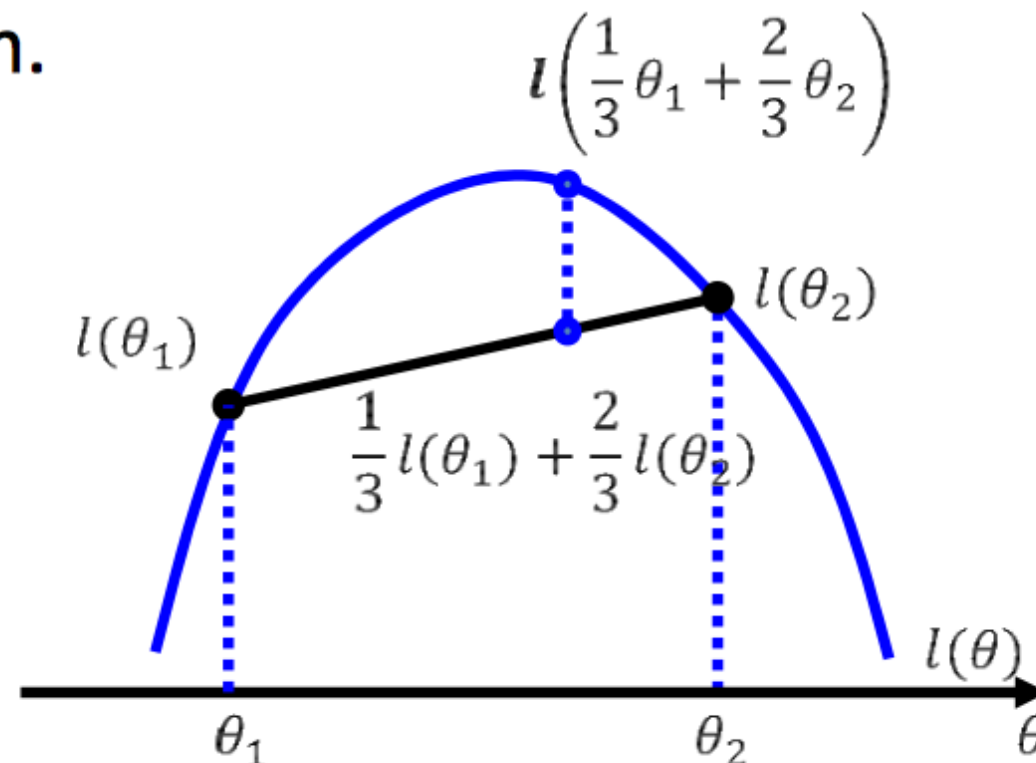
- Setting it to 0 does not lead to closed form solution

The Objective Function

- Find θ , such that the conditional likelihood of the labels is maximized

$$\max_{\theta} l(\theta) := \log \prod_{i=1}^{\bar{n}} p(y^{\{i\}} | x^{\{i\}}, \theta)$$

- Good news: $l(\theta)$ is concave function of θ , and there is a single global optimum.



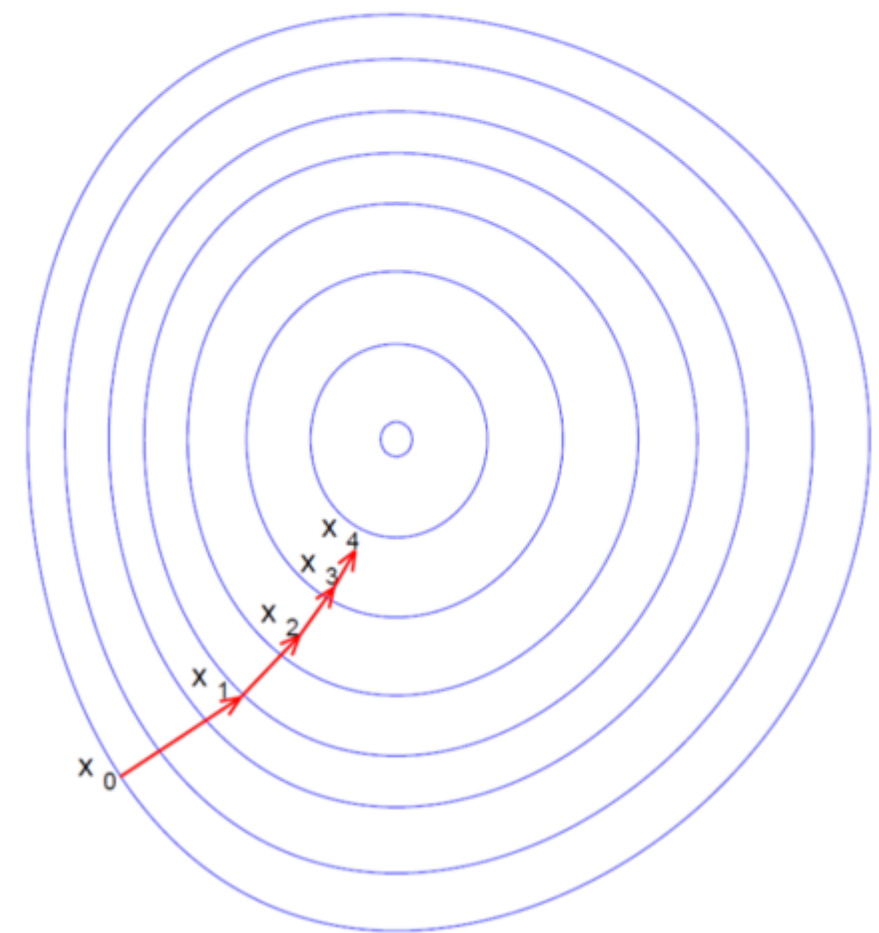
- Bad new: no closed form solution (resort to numerical method)

Gradient Descent

- One way to solve an *unconstrained* optimization problem is gradient descent
- Given an initial guess, we *iteratively* refine the guess by taking the direction of the negative gradient
- Think about going down a hill by taking the steepest direction at each step
- Update rule

$$x_{k+1} = x_k - \gamma_k \nabla f(x_k)$$

γ_k is called the step size or learning rate



Gradient Ascent(concave)/Descent(convex) algorithm

- Initialize parameter θ^0

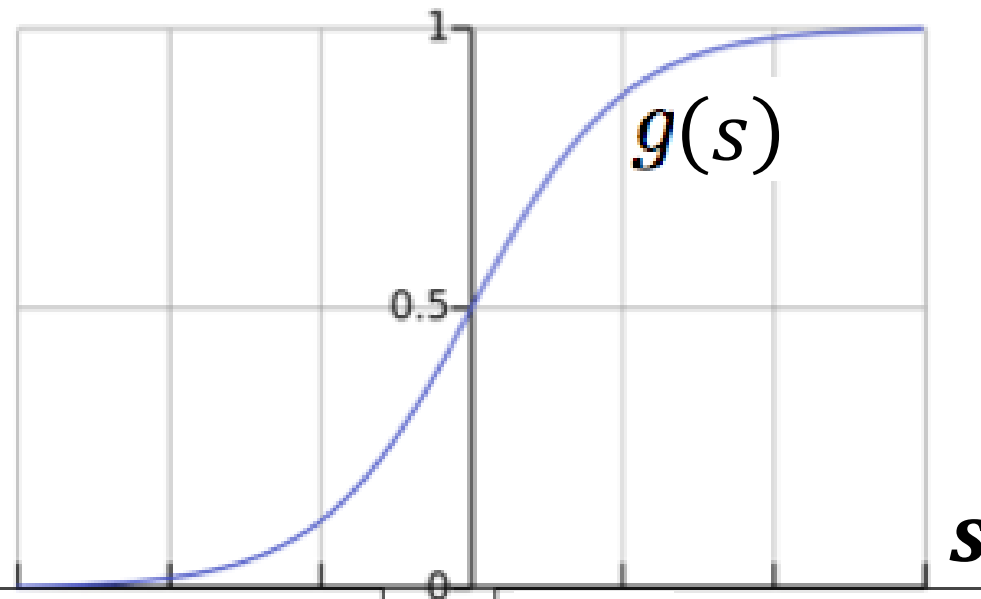
- Do

$$\theta^{t+1} \leftarrow \theta^t + \eta \sum_i (x^{\{i\}})^T (y^{\{i\}} - 1) + (x^{\{i\}})^T \frac{\exp(-x^{\{i\}}\theta)}{1 + \exp(-x^{\{i\}}\theta)}$$

- While the $||\theta^{t+1} - \theta^t|| > \epsilon$

Logistic Regression

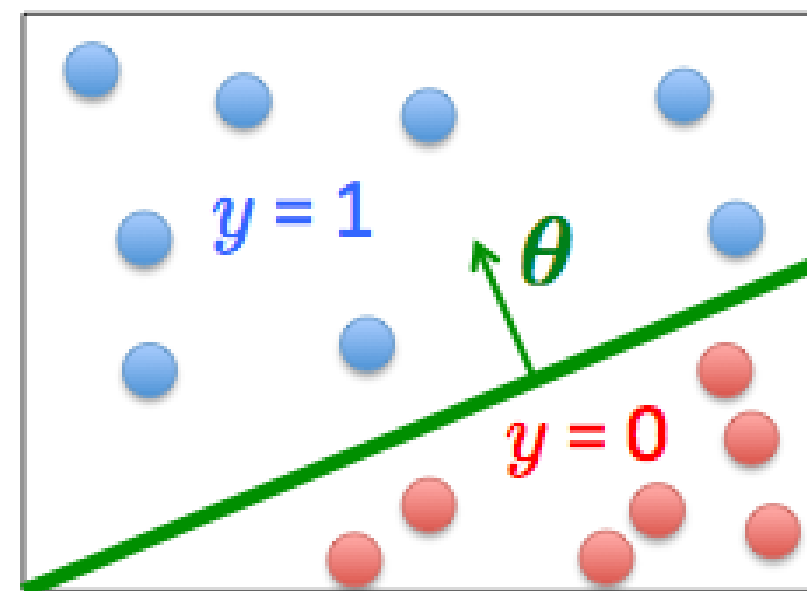
$$g(s) = \frac{e^s}{1 + e^s} = \frac{1}{1 + e^{-s}}$$
$$s = x\theta$$



$x\theta$ should be large negative
values for negative instances

$x\theta$ should be large positive
values for positive instances

- Assume a threshold and...
 - Predict $y = 1$ if $g(s) \geq 0.5$
 - Predict $y = 0$ if $g(s) < 0.5$

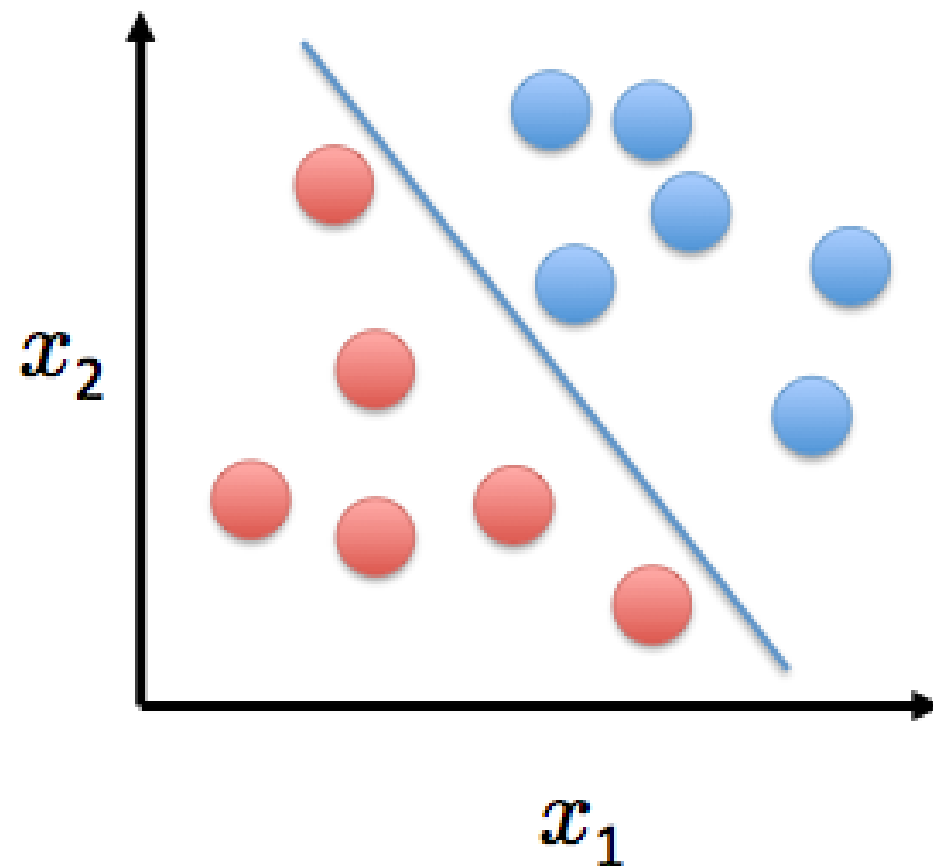


Outline

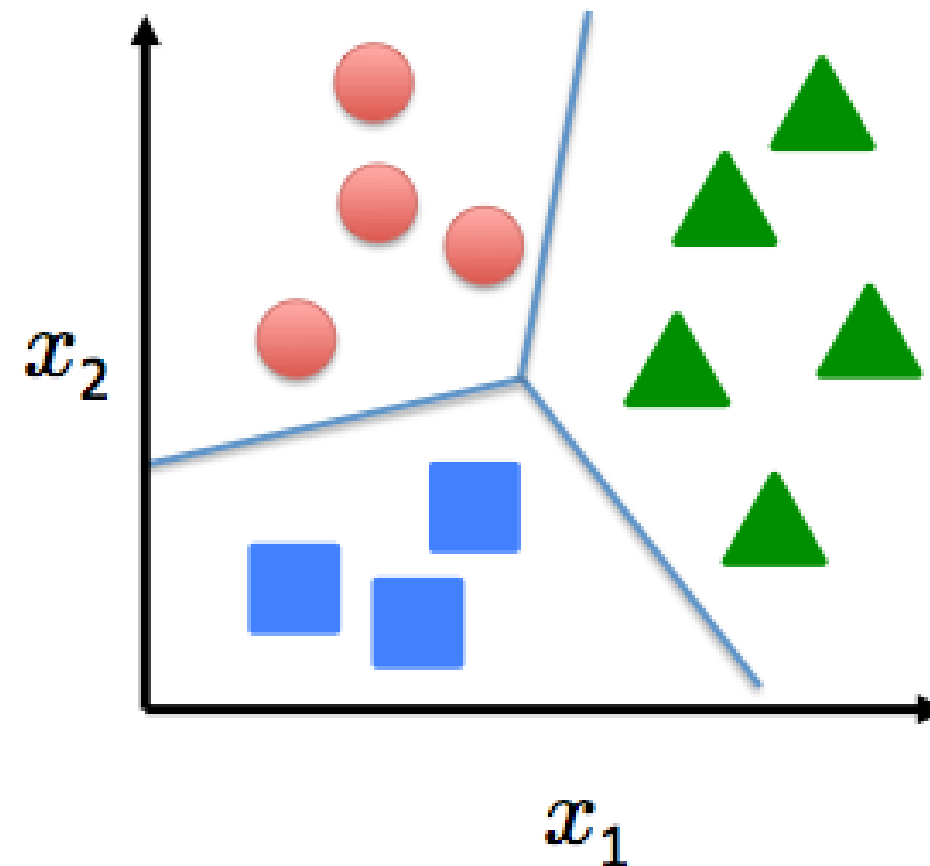
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Multiclass Logistic Regression

Binary classification:



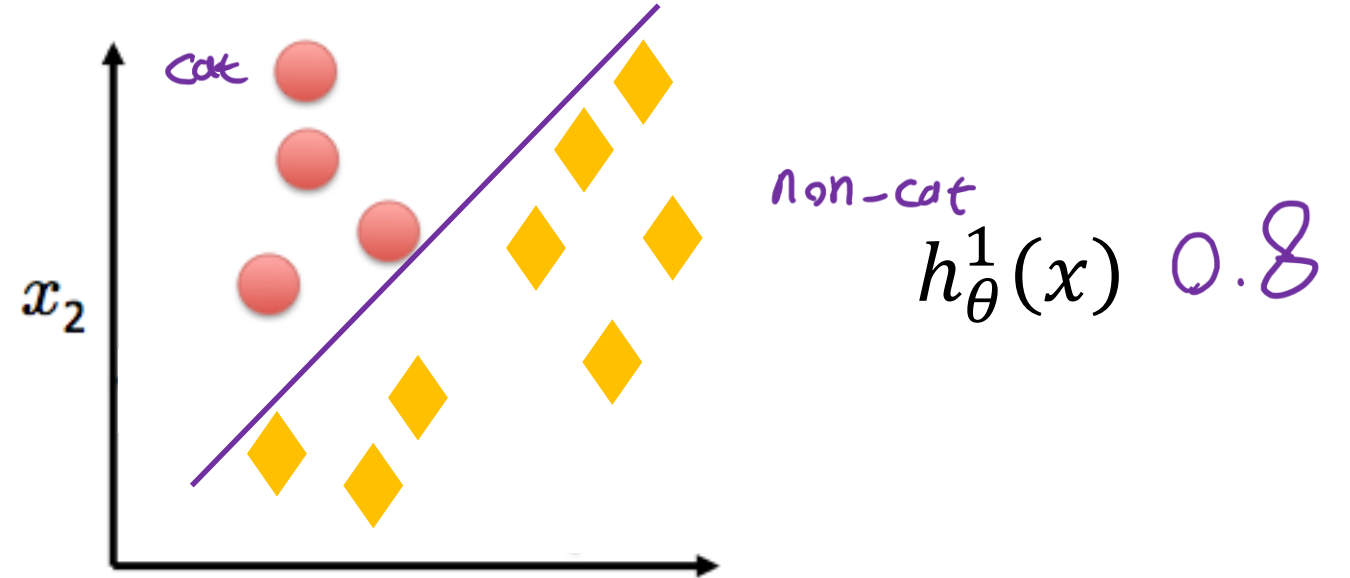
Multi-class classification:



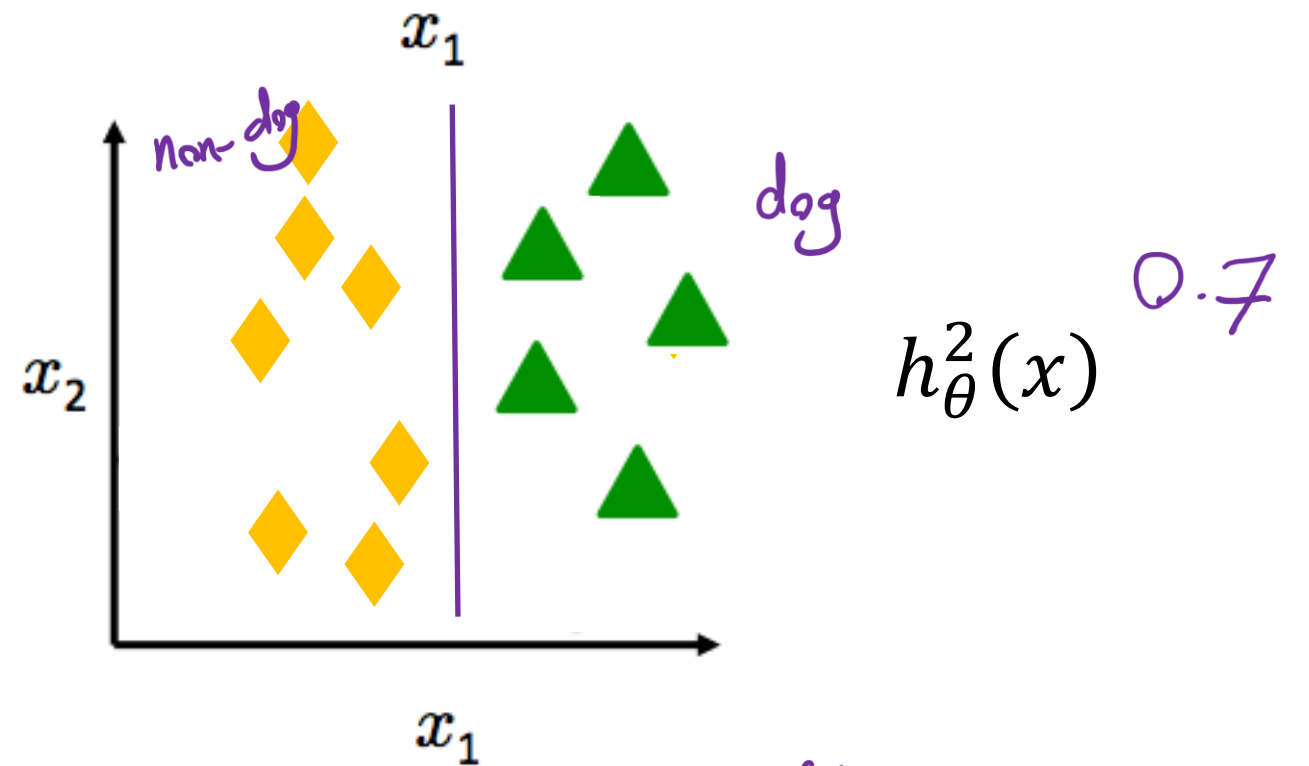
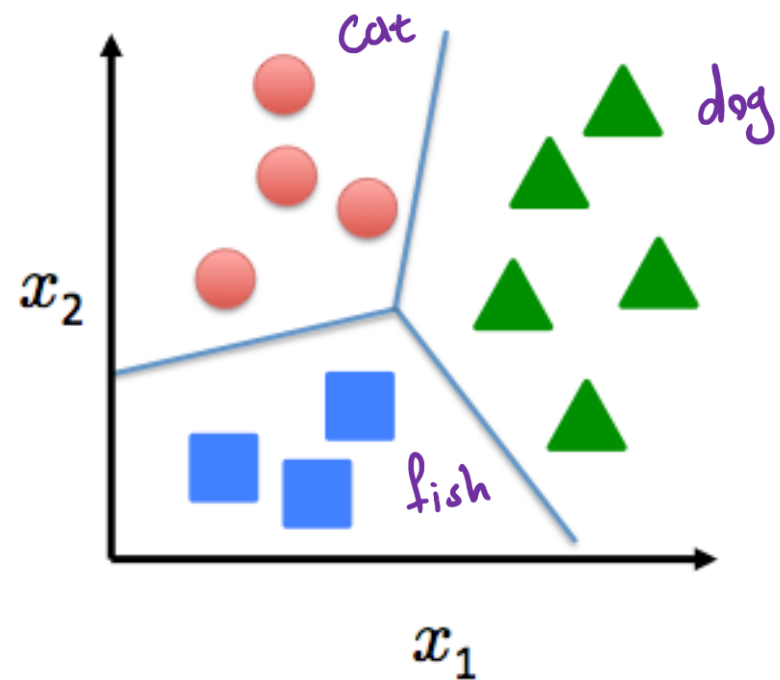
Disease diagnosis: healthy / cold / flu / pneumonia

Object classification: desk / chair / monitor / bookcase

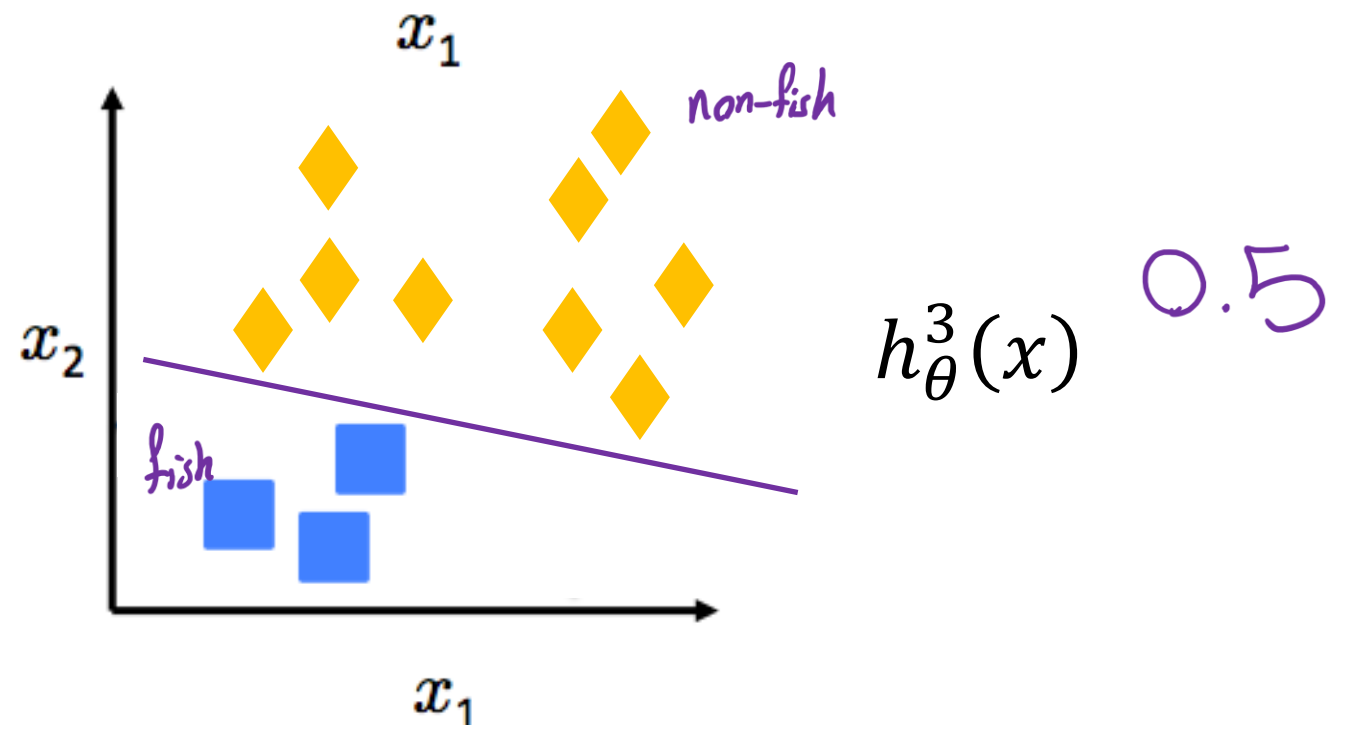
One-vs-all (one-vs-rest)



Multi-class classification:



$$h_{\theta}^{(m)}(x) = p(y = 1|x, \theta) \quad (m = 1, 2, 3)$$



One-vs-all (one-vs-rest)

Train a logistic regression $h_{\theta}^{(m)}(x)$ for each class m

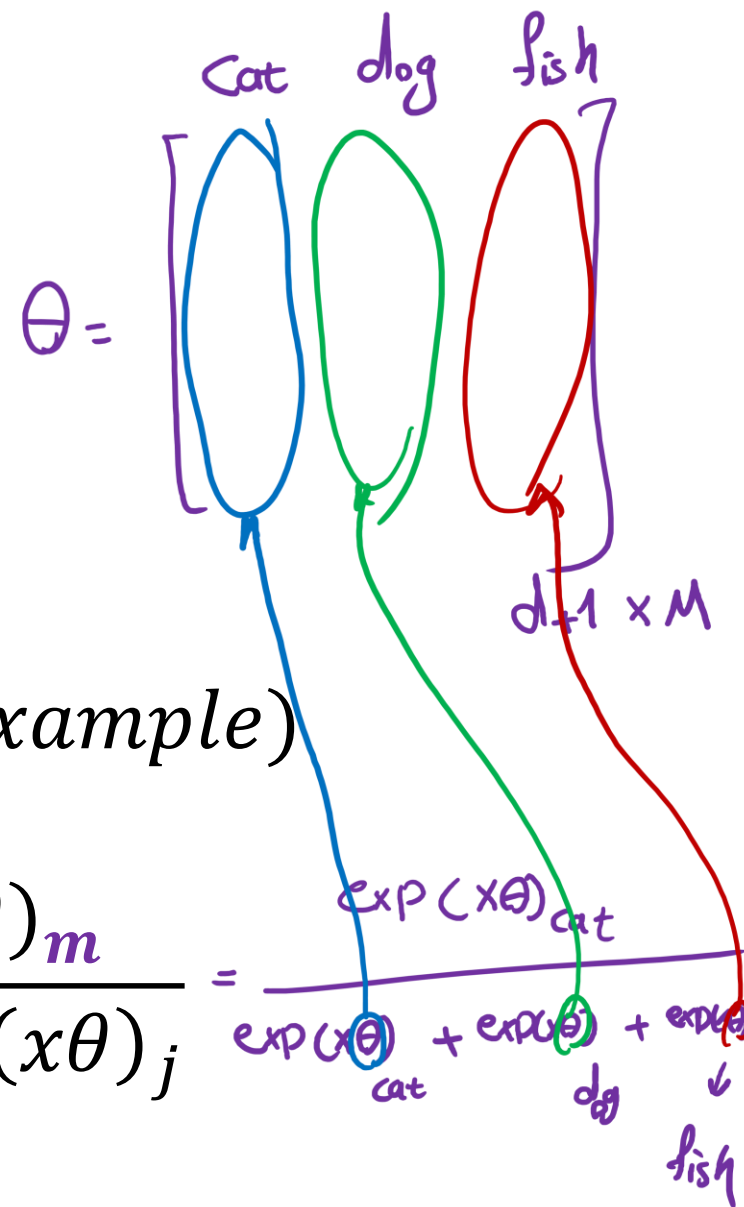
To predict the label of a new input x , pick class m that maximizes:

$$\max_i h_{\theta}^{(m)}(x)$$

$$\Theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_d \end{bmatrix}_{d+1 \times 1}$$

Using Softmax

$$L(\theta) = - \sum_{i=1}^N y_a^{\{i\}} * \log(y_p^{\{i\}})$$



$$y_a = [\text{cat}, \text{dog}, \text{fish}] = [1, 0, 0]$$

$\Rightarrow$ there are M classes ($M = 3$ in this example)

$$y_p \text{ for class } m = \text{softmax}(x\theta) = \frac{\exp(x\theta)_m}{\sum_{j=0}^M \exp(x\theta)_j} = \frac{\exp(x\theta)_{\text{cat}}}{\exp(x\theta)_{\text{cat}} + \exp(x\theta)_{\text{dog}} + \exp(x\theta)_{\text{fish}}}$$

$$y_p = [0.6, 0.3, 0.1]$$

$$SGD \Rightarrow \theta^{t+1} \leftarrow \theta^t - \alpha \nabla L(\theta)$$

$$\theta^{t+1}_{d+1 \times m} \leftarrow \theta^t_{d+1 \times m} - \alpha x^T (y_p - y_a)$$

Take-Home Messages

- Generative and Discriminative Classification
- The Logistic Regression Model
- Understanding the Objective Function
- Gradient Descent for Parameter Learning
- Multiclass Logistic Regression